

Multiple Choice (4 marks)

- Statement 1 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)+g(x)) = \max(\deg f(x), \deg g(x))$. Statement 2 | If F is a field then every...
 - True, True
 - False, False
 - True, False
 - False, True
- Find the sum of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$.
 - $2x^2 + 5$
 - $6x^2 + 4x + 6$
 - 0
 - $x^2 + 1$

True / False (16 marks)

Answer True or False and justify briefly.

- True or False: The correct answer to 'Let C be the circle in the yz -plane whose equation is $(y - 3)^2 + z^2 = 1$. If C is revolved around the z -axis, the surface generated is a...' is ' $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$ '.
- True or False: The correct answer to 'Find the order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15})/(\langle 1, 1 \rangle)$ ' is '1'.
- True or False: The correct answer to 'Statement 1 | For any two groups G and G' , there exists a homomorphism of G into G' . Statement 2 | Every homomorphism is a one-to-one map.' is 'False, False'.
- True or False: The correct answer to 'Using Fermat's theorem, find the remainder of 3^{47} when it is divided by 23.' is '4'.
- True or False: The correct answer to 'Determine whether the polynomial in $\mathbb{Z}[x]$ satisfies an Eisenstein criterion for irreducibility over $\mathbb{Q}$. $8x^3 + 6x^2 - 9x + 24$ ' is 'Yes, with $p=3$ '.
- True or False: The correct answer to 'What is the area of an equilateral triangle whose inscribed circle has radius 2?' is ' $12\sqrt{3}$ '.

7. True or False: The correct answer to 'A tree is a connected graph with no cycles. How many nonisomorphic trees with 5 vertices exist?' is '3'.
8. True or False: The correct answer to 'Statement 1 | If H is a subgroup of G and a belongs to G then $|aH| = |Ha|$. Statement 2 | If H is a subgroup of G and a and b belong to G ,...' is 'False, True'.

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. Expand the following expression: $3(8x^2 - 2x + 1)$.

End of Paper